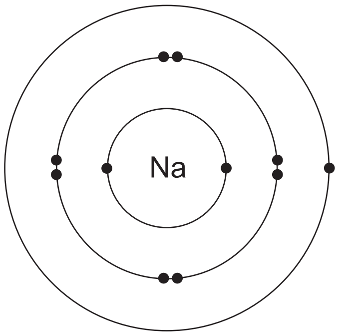
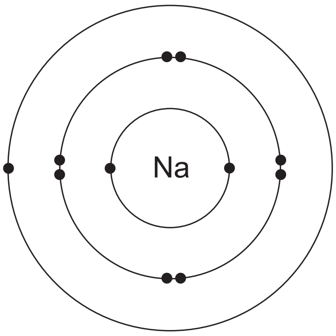
Examiner only

**4.** (a) Sodium reacts with oxygen to form the ionic compound sodium oxide.

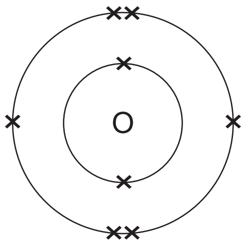
In the following diagrams, dots and crosses represent electrons in sodium and oxygen atoms.




sodium atom



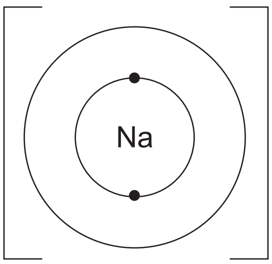

sodium atom



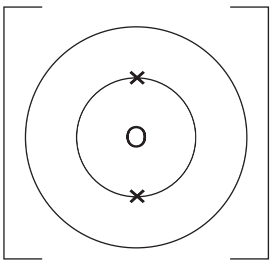

oxygen atom

(i) Complete the diagrams below.

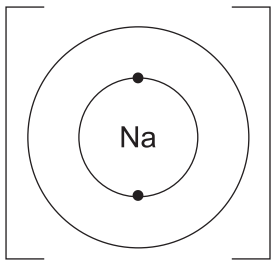
- Complete the electronic structures of the sodium ions and the oxide ion formed.
- Give the charges on the sodium and oxide ions. [3]




sodium ion




oxide ion




sodium ion

(ii) Complete and balance the symbol equation for the reaction. [2]

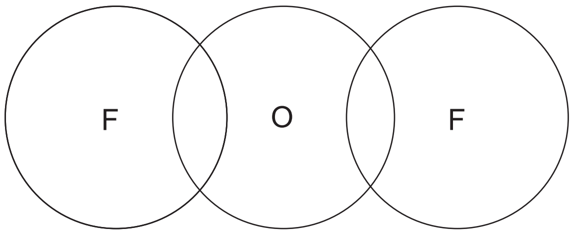
$4Na \quad + \quad O_2 \longrightarrow \boxed{\phantom{00}}$ .......................................

Examiner only

(b) (i) The table shows the electronic structure of the elements present in oxygen difluoride, $F_2O$.

| Element | Electronic structure |
|---|---|
| fluorine | 2,7 |
| oxygen | 2,6 |

Complete a dot and cross diagram to show the bonding in a molecule of oxygen difluoride. [2]

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(ii) Tick (✓) the box next to the statement that explains why oxygen difluoride has low melting and boiling points. [1]

bonds within the molecules are strong ☐

bonds within the molecules are weak ☐

bonds between the molecules are strong ☐

bonds between the molecules are weak ☐

8



 